Problem 5.1

Proof. The proof is by induction. Let $P(n)$ be the proposition that there for all n ≥ 1, the set with the length of n always exist an mininum element.
Base case: When $n = 1$, the set contain only one real number. Naturally, this single element is the minimum element in the set. Thus $p(1)$ is true.

Inductive hypothesis: Assume that $p(k)$ is true for an arbitrary integer $k \geq 1$. That is, any set of real number with k element has a minimum element.

Inductive step: We want to prove that $P(k+1)$ holds. Let S be a arbitrary set with $k + 1$ element.

We can remove an element, says x , from S. The remaining set $S' = S \setminus x$ has exactly k element. By the inductive hypothesis, S' must have a minimum, which we call m.

Now we compare x with m:

- If $x < m$, then x is smaller than all element in S', so x is the minimum of S.
- If $x \geq m$, then m is smaller than or equal to all element in the S, so m is the minimum of S.

In either case, the set S with $k + 1$ element has a minimum element. This proves that $p(k + 1)$ is true.

So wo proved $p(n)$ is true for all $n \geq 1$.

$\square$